

Eric Zmitrovich

Software Engineer

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Computer Science graduate with a focus in AI and Machine Learning and a minor in Mathematics. Experienced in AI/ML systems, iOS/Android mobile development, and embedded programming. Shipped production apps, implemented custom neural networks in C, and built ML models using TensorFlow/PyTorch.

EDUCATION

Portland State University

B.S. in Computer Science with minor in Mathematics

M.S. in Computer Science with focus on AI & Machine Learning

Portland, Oregon

2018-2022 | 3.90 GPA

2022-2024 | 3.84 GPA

TECHNICAL SKILLS

Languages: C/C++, Python, Java, Swift, TypeScript, Go, Rust, SQL

Frameworks/Tools: Git, OpenAI API, TensorFlow, PyTorch, Supabase

Platforms: Linux, MacOS, Windows, AWS

Core Competencies: AI & Machine Learning, Mobile App Development, Embedded Programming

TECHNICAL EXPERIENCE & PROJECTS

Bigfoot Brain — iOS App (*Xcode, Swift, OpenAI API, Supabase, TypeScript*) | [App Store](#)

- Developed and launched my own AI-powered text rewriter & translator app, live on the App Store.
- Integrated OpenAI API through Supabase edge functions to deliver real-time language processing.

Push-to-Mumble — VOIP Android App (*Java, Git, Open-source Library*) | [YouTube](#)

- 6-person capstone project in collaboration with Datalogic
- Built a walkie-talkie app for warehouse scanners by extending open-source Mumla client.
- Delivered push-to-talk functionality on enterprise Android scanners, improving warehouse communication.

The Avanth Stone — 2D Adventure Game (*Python, Pygame, Game Design*) | [GitHub](#) | [YouTube](#)

- Developed a small Zelda-style 2D game with animated sprites, character movement, and combat logic.
- Designed game map, integrated public-domain audio assets, and implemented core gameplay mechanics.

Encrypted Messenger — Secure Messaging Client (*Python, Cryptography, Double Ratchet*) | [GitHub](#)

- Built a client-server-client messenger with Double-Ratchet encryption.

Deep Reinforcement Learning Agent — Atari Games Agent (*Java, Git, Open-source Library*) | [GitHub](#)

- Designed a multithreaded neural network in C (POSIX threads) to train reinforcement learning agents.
- Benchmarked custom implementation vs. TensorFlow across multiple OpenAI Gym environments.

ASL Image Classifier — ML Model (*Python, TensorFlow, Scikit-learn*) | [GitHub](#)

- Built and trained a neural network to classify American Sign Language hand gestures with 95% accuracy.

RELEVANT COURSEWORK

Computer Science: Data Structures & Algorithms, Operating Systems, Computer Hardware, AI & Machine Learning, Computer Vision, Reinforcement Learning, Large Language Models, Data Clustering, Adv. Java Programming, Database Management, Embedded Rust Programming, Rust Web Development, Computer Sound and Audio, Computer Graphics, Cryptography

Mathematics: Calculus 1-4, Linear Algebra, Differential Equations, Discrete Mathematics, Group Theory, Agent-Based Scientific Modeling, Theory of Computation